

DESTINATION EARTH



THE CLIMATE CHANGE ADAPTATION DIGITAL TWIN

Outi Sievi-Korte, Theresa Kiszler, Jenni Kontkanen, Matteo Nurisso, Brynjar Sigurdsson, Bruno de Paula Kinoshita, Pierre-Antoine Bretonniere, Devaraju Narayanappa, Mario Acosta, Suraj Polade, Thomas Jung, Daniel Klocke, Francisco Doblas-Reyes, Nikolay Koldunov, Aina Gaya-Àvila, Jost von Hardenberg, Paolo Davini, Barbara Früh, Stephan Thober, Sebastian Milinski, and Francesc Roura Adserias and the Climate DT team

CSC – IT Center for Science, Finland, outi.sievi-korte@csc.fi



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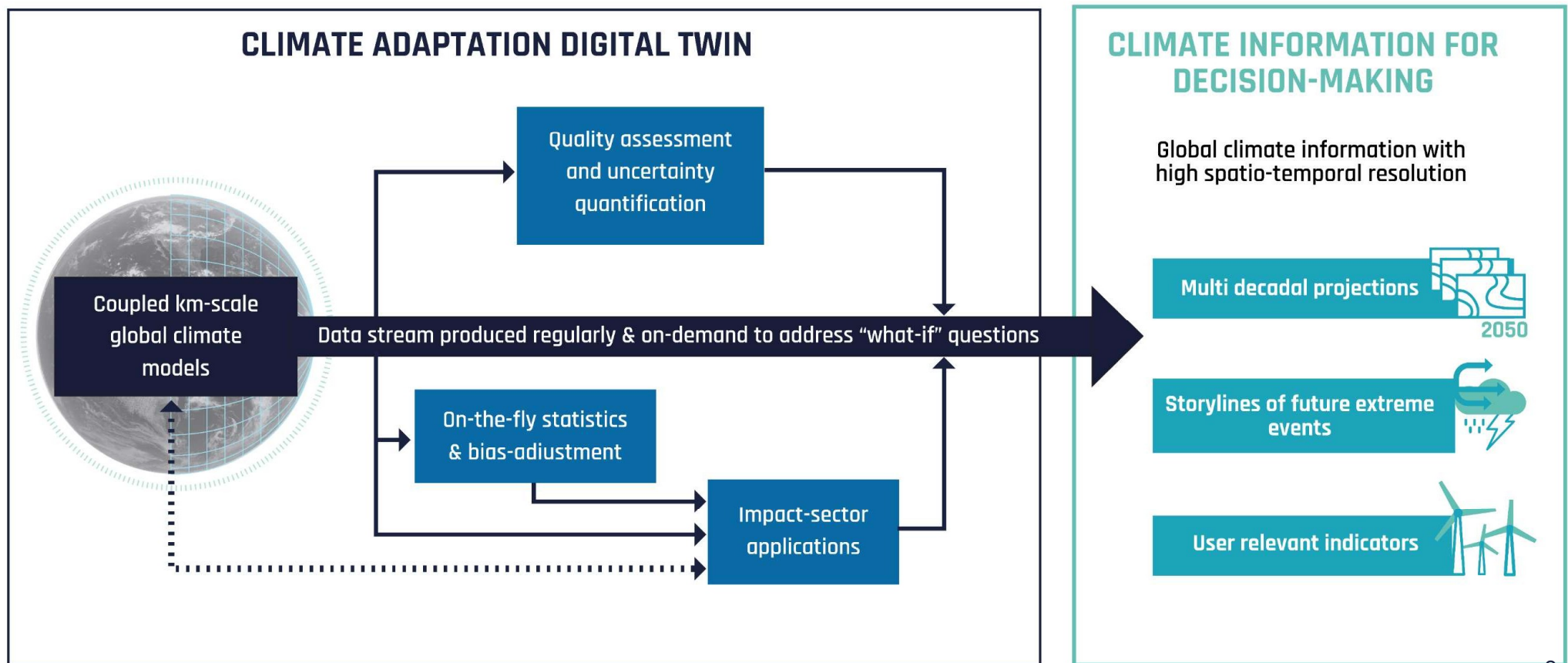
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CLIMATE DT PRODUCES ACTIONABLE INFORMATION





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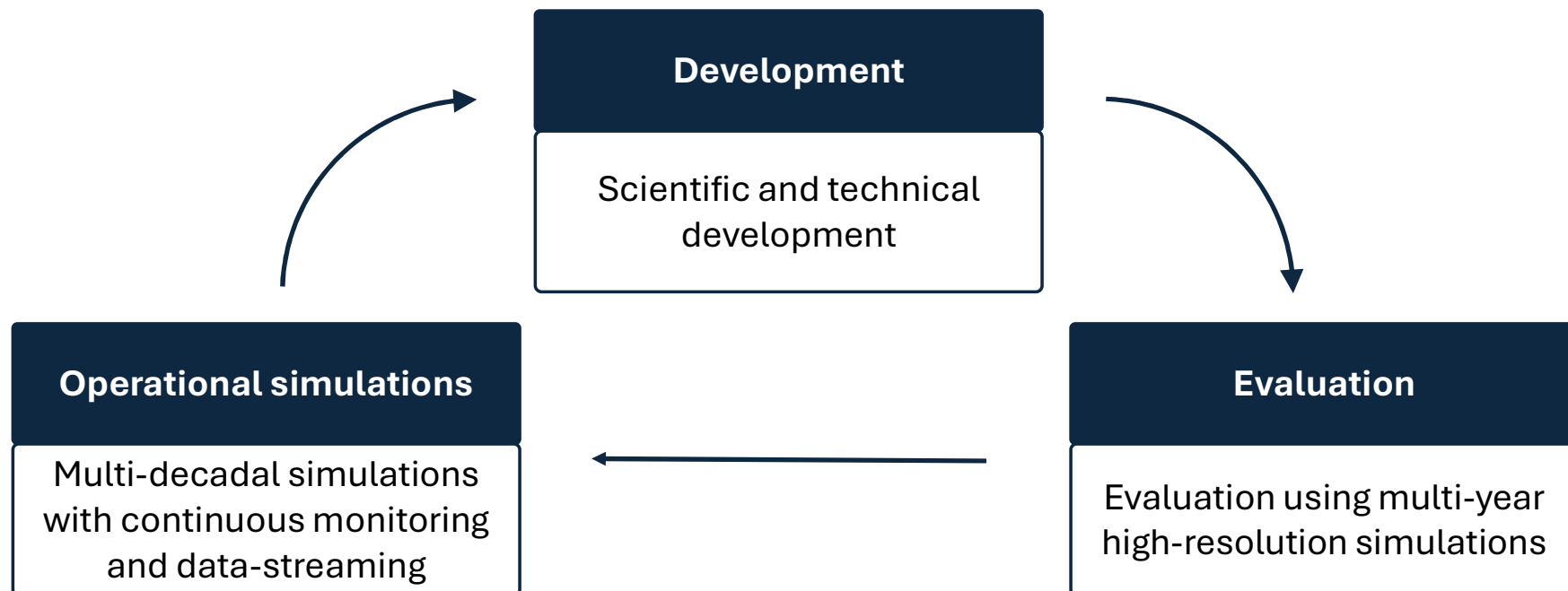
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OPERATIONAL CAPABILITIES ENABLE REGULAR PRODUCTION OF NEW DATA

Climate DT operational cycle includes multiple steps for development, evaluation and operationilization





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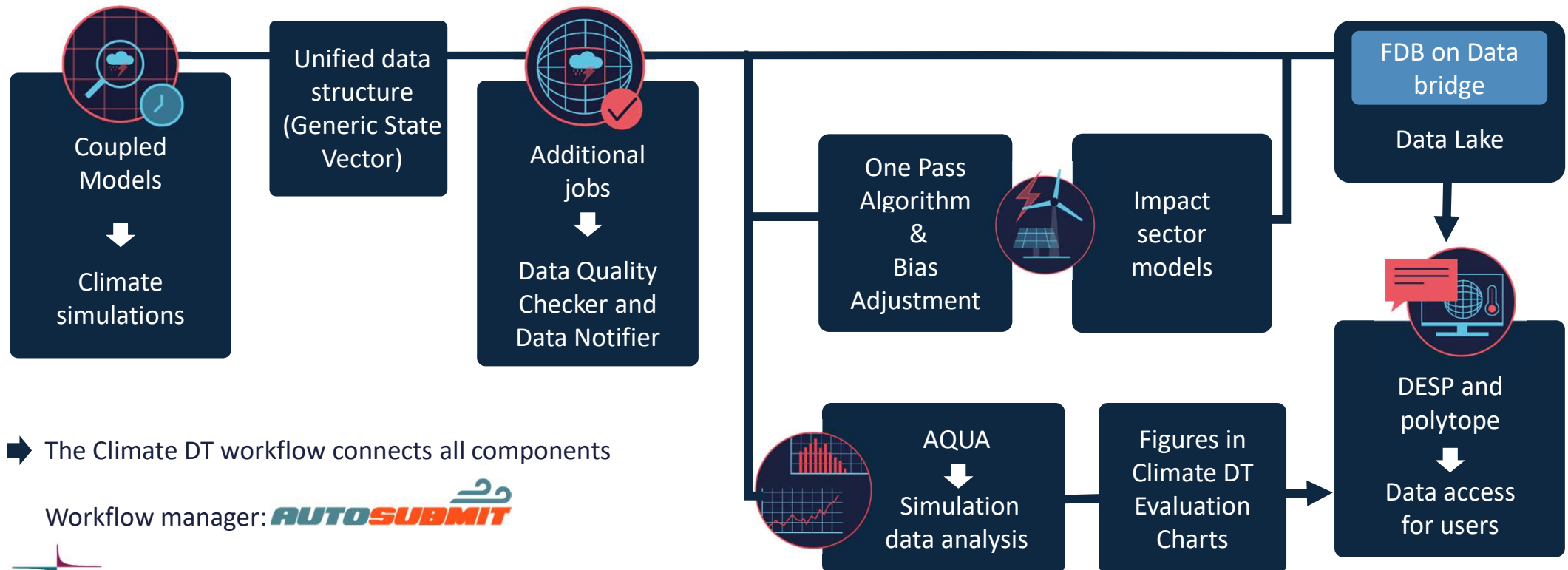
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The Climate DT Workflow

Fields DataBase (FDB) on EuroHPC



➡ The Climate DT workflow connects all components

Workflow manager: **AUTOSUBMIT**





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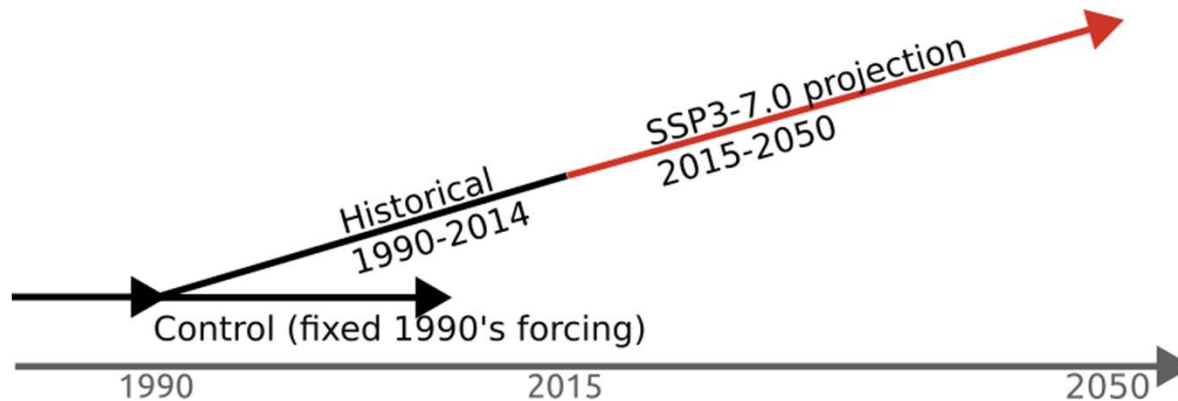
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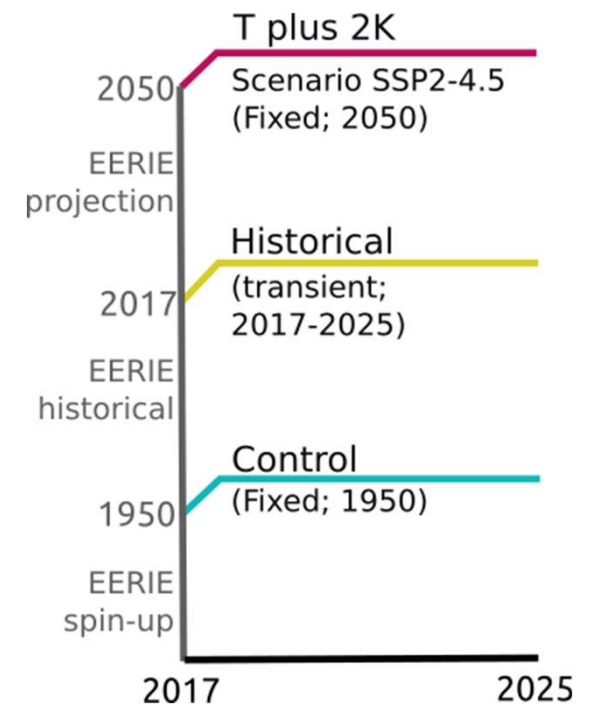
Generation 2 Simulations

- Coupled ocean and atmosphere
- IFS-FESOM, IFS-NEMO, ICON
- Using CMIP6 forcings

Multi-decadal simulations 5km with all models



Storylines with IFS-FESOM 10km





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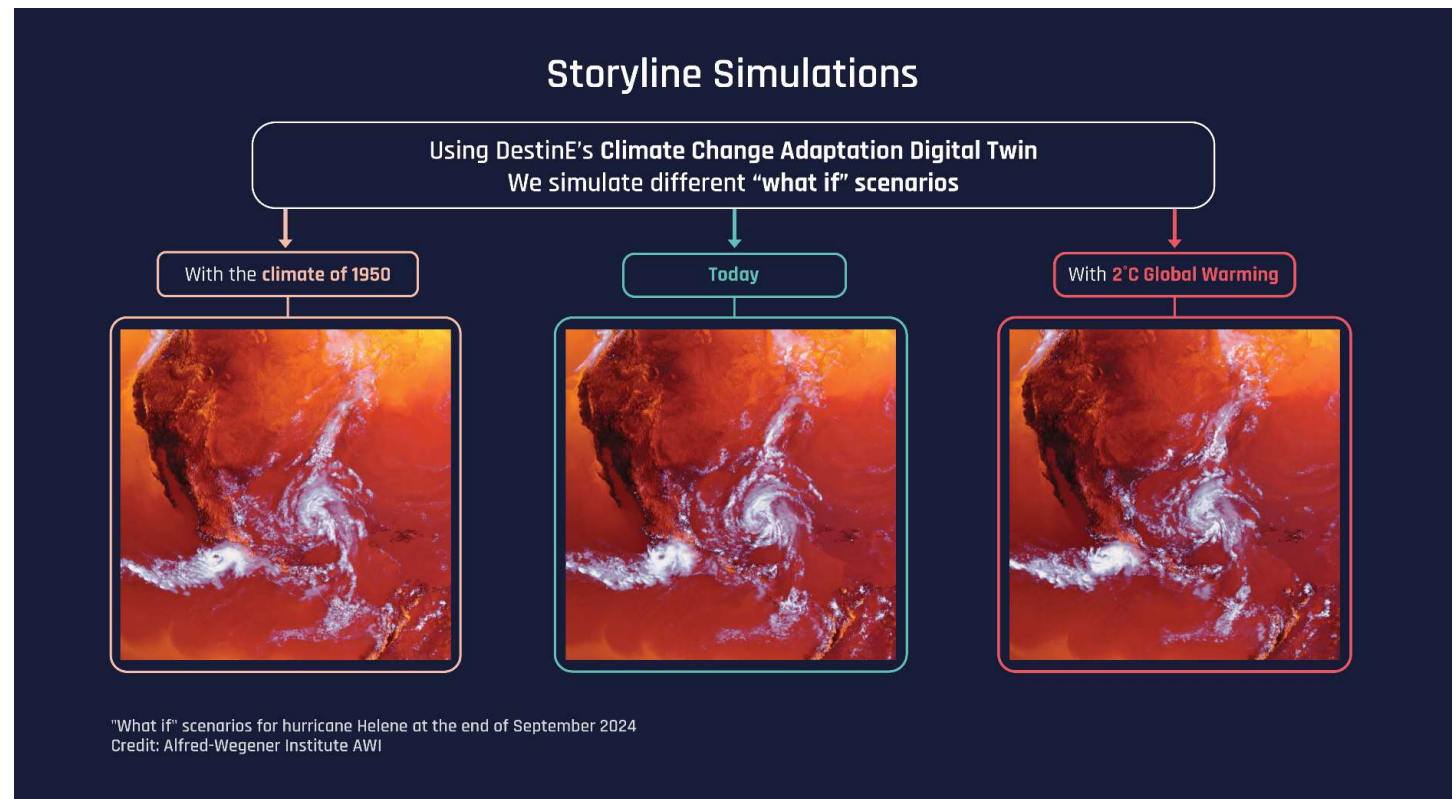
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Generation 2 Storyline Simulations

- Simulations with IFS-FESOM
- 10km resolution, 2017-2025 (will be extended)
- Uses spectral nudging to constrain the large-scale dynamics*
- Ability to replay extreme weather under different “what-if” scenarios





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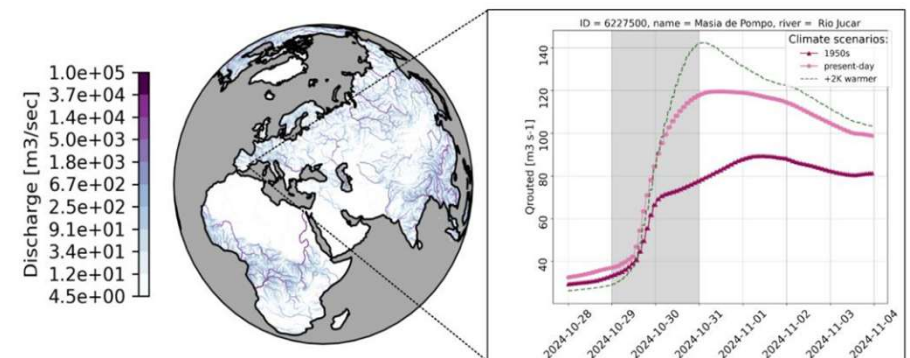
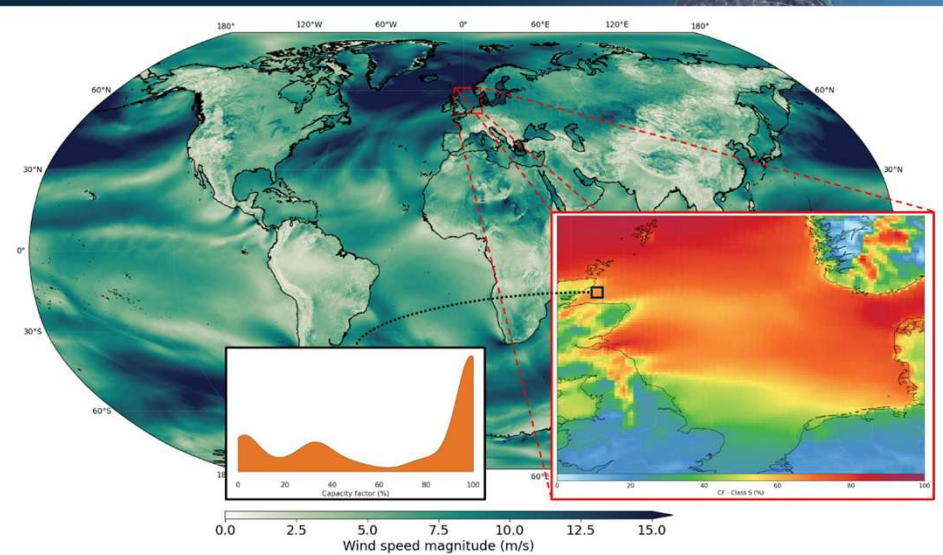


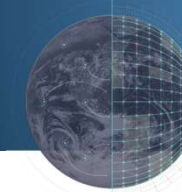
Impact sector models

- Create information for public institutions, start-ups and SMEs, decision makers

Current impact sector models:

- Wildfires: Modelling wildfires risk in the future
- HydroLand and HydroMet: Provide statistics for flooding, runoff and groundwater
- Energy indicators: Provide analysis of renewable energy capacities





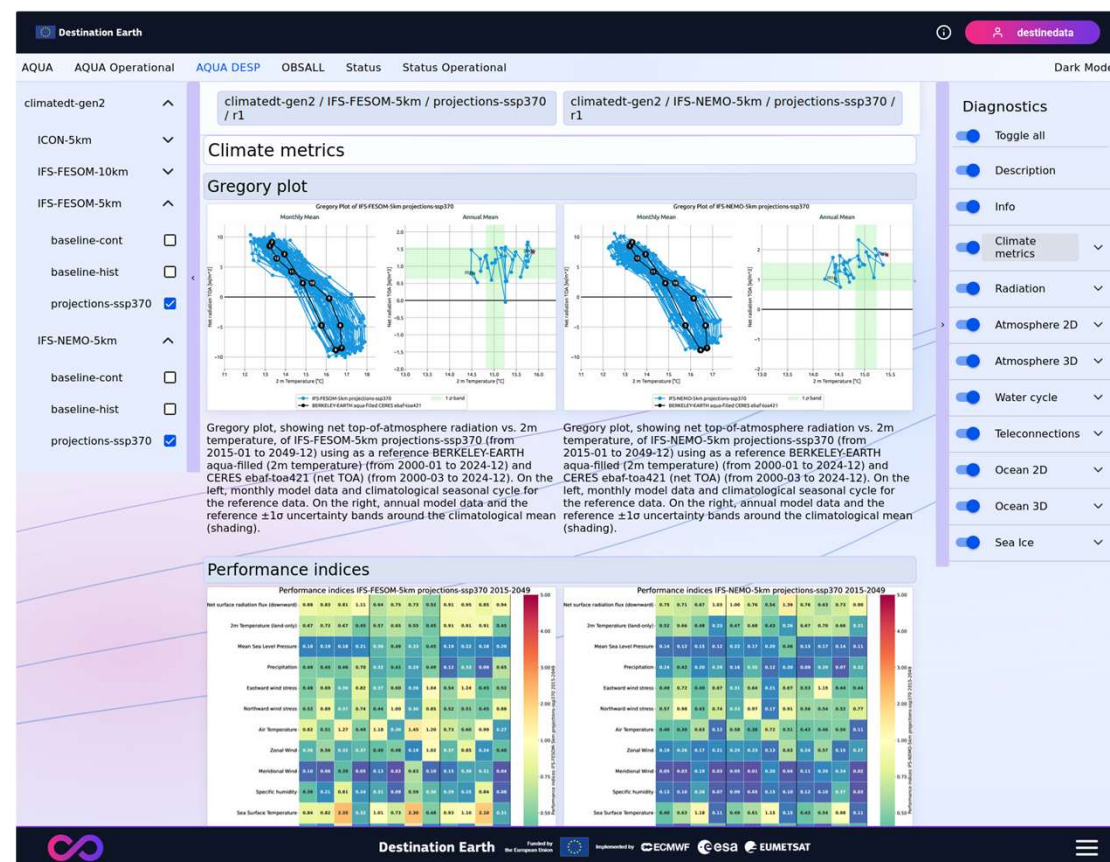
Monitoring and Evaluation

Open-source software AQUA



- Creates a low-resolution archive
- Real-time monitoring capabilities
- Provides model evaluation figures

Example diagnostics: Tropical Rainfall, Tropical Cyclones detection, Timeseries and biases, Radiation budget, Sea ice extent, Teleconnection indices, and more



The Climate DT Evaluation Charts for real-time monitoring.



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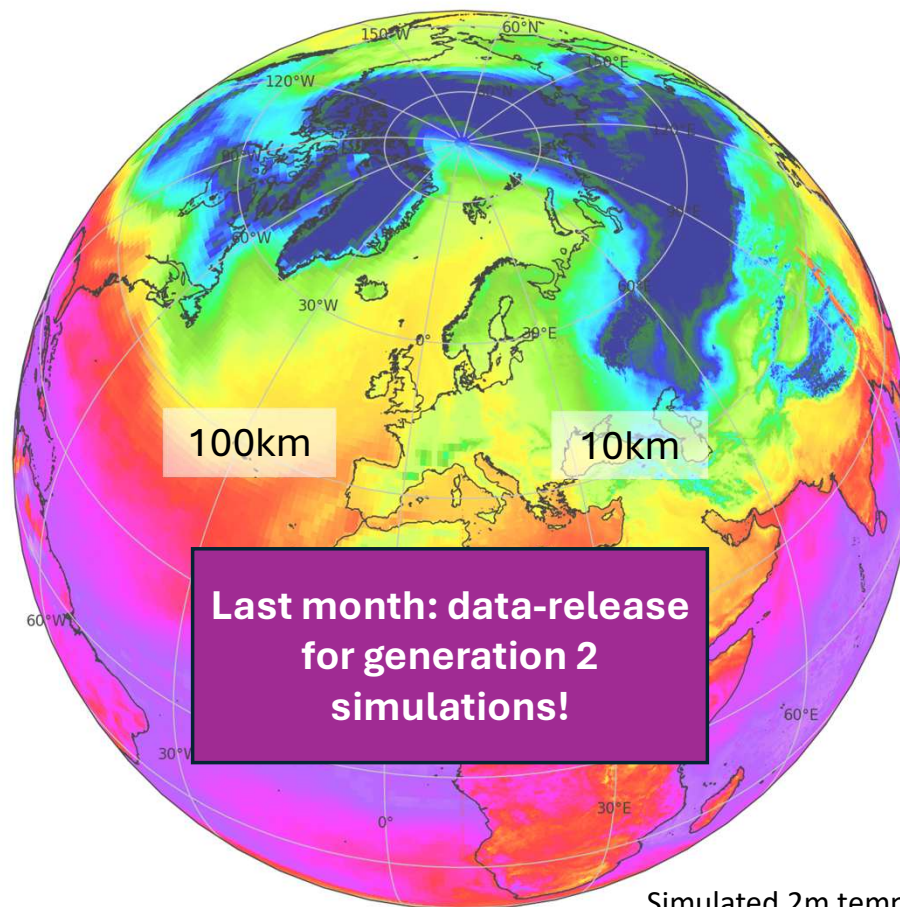
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Explore Climate DT

New article in GMD:

Doblas-Reyes, Kontkanen
et al., The Destination
Earth digital twin for
climate change
adaptation, 2026



Website:

<https://platform.destine.eu/climate-dt/>

User guide:

- Overview of models and simulations
- Impact sector models explanation
- Introduction to data and access

We acknowledge the EuroHPC Joint Undertaking (JU) for awarding this project access to the EuroHPC supercomputer LUMI and MareNostrum5 through a EuroHPC JU Special Access call.